

PostureIQ: AI-Powered Fitness Analysis Platform

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Abstract

PostureIQ is an AI-powered fitness analysis platform developed to evaluate exercise posture using computer vision and machine learning. The system processes uploaded workout videos, extracts human pose landmarks through MediaPipe Pose, and performs exercise-specific posture assessment using trained machine learning models.

The platform provides posture accuracy scores, repetition counts, annotated output videos, and workout analytics that help users identify movement errors and improve exercise execution. Dedicated analysis pipelines have been developed for Bicep Curls, Squats, Lunges, and Planks.

By integrating computer vision, machine learning, and full-stack web technologies, PostureIQ aims to make intelligent fitness guidance more accessible while helping users reduce exercise-related injuries and improve training quality.

Keywords: Computer Vision, Machine Learning, Fitness Analytics, Pose Estimation, MediaPipe, Random Forest, FastAPI, Next.js, Exercise Analysis

1. INTRODUCTION

Proper exercise posture plays a critical role in maximizing workout effectiveness and reducing the risk of long-term injuries. While online fitness content has become widely accessible, many individuals continue to perform exercises without professional supervision, making it difficult to identify posture mistakes and movement inefficiencies.

Incorrect exercise execution can reduce muscle engagement, limit training effectiveness, and significantly increase the likelihood of chronic injuries. Most fitness applications primarily focus on workout planning, activity tracking, and exercise demonstrations, offering limited support for posture evaluation and movement-quality assessment.

To address this challenge, PostureIQ was developed as an AI-powered fitness analysis platform that combines computer vision, machine learning, and modern web technologies to automatically evaluate exercise form. The system analyzes uploaded workout videos, extracts body landmarks, evaluates posture quality, counts repetitions, and generates annotated visual feedback for users.

The platform is designed to provide an accessible and scalable alternative to traditional fitness supervision by enabling users to receive objective posture analysis directly from recorded workout sessions.

2. EXISTING SYSTEMS AND OBSERVATIONS

Several fitness and exercise-tracking platforms currently exist in the market. Most focus on workout recommendations, activity monitoring, or general fitness tracking rather than detailed posture assessment.

Common examples include:

- Fitness tracking applications
- Smartwatch-based activity monitors
- Online workout coaching platforms
- Exercise demonstration systems

Although these solutions provide useful functionality, several limitations remain:

- Limited posture analysis capabilities
- Lack of exercise-specific corrective feedback
- Dependence on professional trainers
- Minimal movement-quality assessment
- Absence of automated posture correction systems

Our analysis indicated that an effective fitness intelligence platform must integrate:

- Computer Vision
- Pose Estimation
- Machine Learning
- Repetition Counting
- Workout Analytics
- User Progress Tracking

PostureIQ was developed to unify these capabilities within a single AI-powered platform.

3. PROPOSED SYSTEM

PostureIQ is proposed as a full-stack fitness intelligence platform capable of evaluating exercise posture through computer vision and machine learning techniques.

The system performs the following core functions:

1. Exercise video acquisition
2. Pose landmark extraction
3. Feature generation
4. Exercise-specific posture classification
5. Repetition counting
6. Workout analytics generation
7. Annotated output video creation

Unlike conventional fitness-tracking applications, PostureIQ focuses on movement quality rather than only workout quantity. The platform enables users to receive detailed posture feedback, performance metrics, and visual analysis reports through a simple video-upload workflow.

4. PROBLEM STATEMENT AND MOTIVATION

Millions of individuals perform workouts at home, in hostels, or in gyms without professional supervision. While exercise tutorials and workout programs are widely available, users often struggle to determine whether they are performing exercises with correct posture and movement patterns.

Improper exercise form can lead to:

- Reduced workout effectiveness
- Poor muscle activation
- Incorrect movement habits
- Increased risk of long-term injuries
- Lack of measurable performance feedback

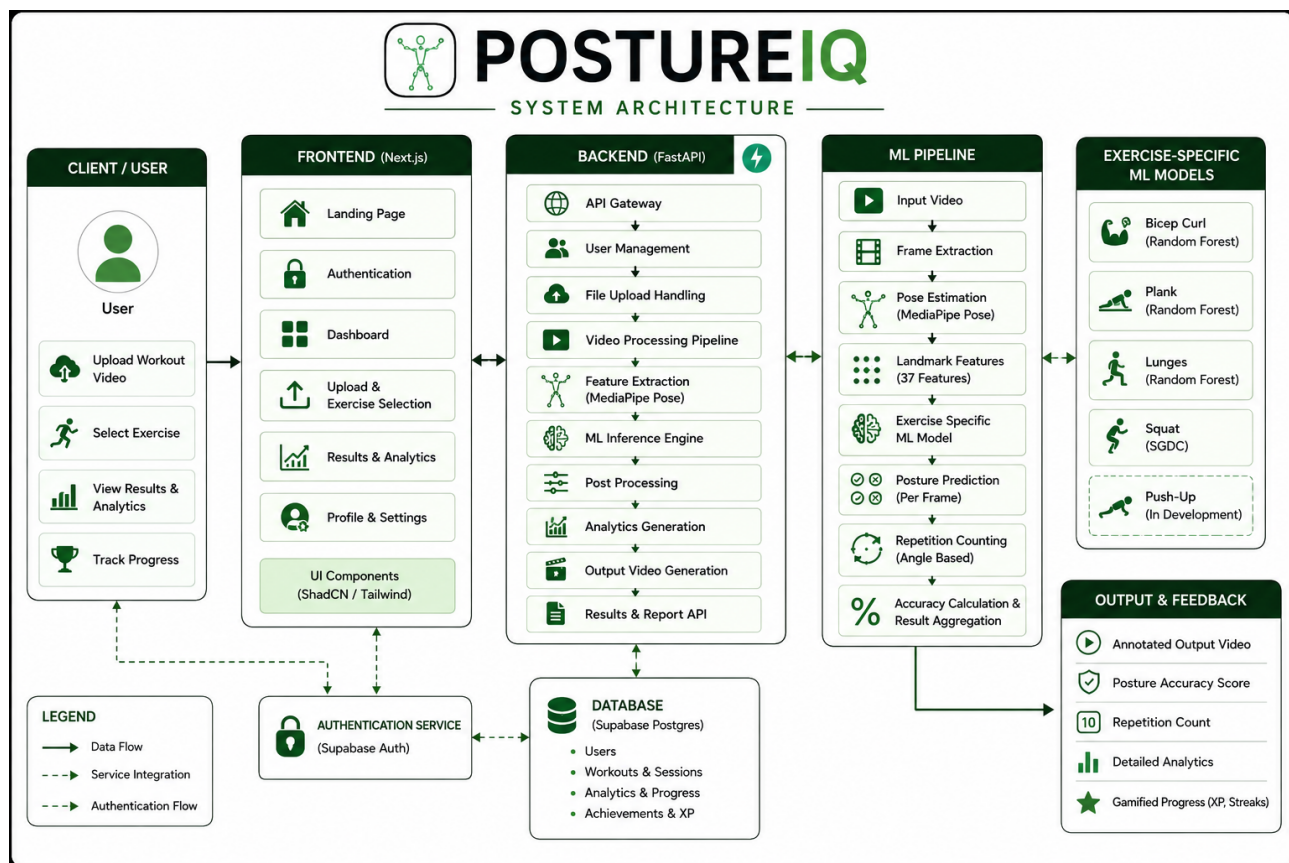


Figure 1. High-Level Overview of the PostureIQ Exercise Analysis Platform

Traditional fitness applications primarily focus on workout planning and tracking rather than detailed movement analysis. Consequently, users receive little to no guidance regarding posture quality during exercise execution.

The motivation behind PostureIQ is to bridge this gap through an intelligent system capable of automatically analyzing workout videos, evaluating posture, tracking repetitions, and providing actionable feedback.

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6. Workout analytics generation
7. Visual feedback through annotated videos

Unlike traditional workout tracking applications, PostureIQ focuses on movement quality rather than only workout quantity.

7. DEVELOPMENT TEAM

The project was developed by a two-member team.

7.1. Mradul Bhartiya

Primary Responsibilities:

- Machine Learning Pipeline Development
- Dataset Collection and Annotation
- Feature Engineering

- Model Training and Evaluation
- FastAPI Backend Development
- Inference Pipeline Design
- Computer Vision Integration

7.2. Moksh Kasture

Primary Responsibilities:

- Frontend Development
- User Interface Design
- Dashboard Development
- Authentication Integration
- User Experience Optimization
- Platform Visualization

8. SYSTEM FEATURES

8.1. Core Features

- Workout Video Upload
- Exercise-Specific Analysis
- Pose Landmark Extraction
- Posture Evaluation
- Repetition Counting
- Annotated Output Video Generation
- Session Analysis Reporting

8.2. User Features

- Home Page
- Authentication System
- Personalized Dashboard
- Exercise Selection Interface
- Workout History
- Profile Management
- Session Analytics

8.3. Gamification Features

- XP Progression System
- Achievement Tiers
- Workout Streak Tracking
- Session Milestones
- Performance Monitoring

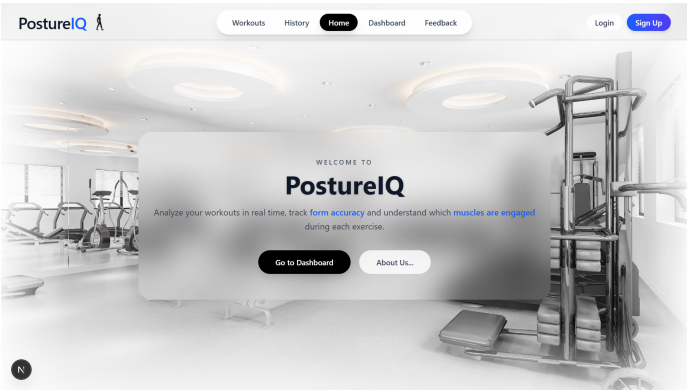


Figure 2. PostureIQ Home Page

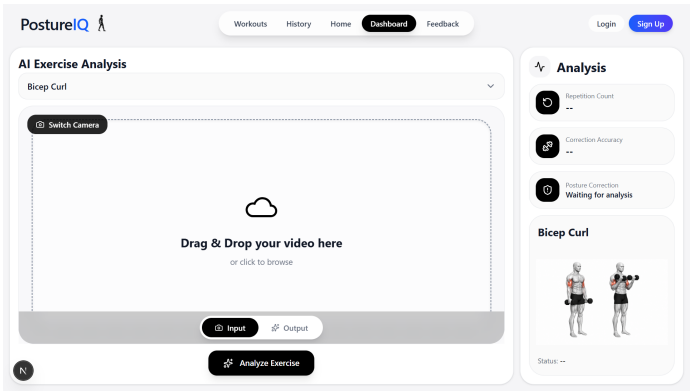


Figure 3. User Dashboard and Analytics Interface

9. TECHNOLOGY STACK

PostureIQ integrates modern web technologies, computer vision frameworks, machine learning libraries, and backend services to deliver a complete fitness analysis platform.

9.1. Frontend Technologies

The frontend application was developed using:

- Next.js
- TypeScript
- Component-Based UI Architecture
- Responsive Dashboard Design

The frontend layer is responsible for workout uploads, analytics visualization, dashboard management, and user interaction.

9.2. Backend Technologies

The backend infrastructure was developed using:

- FastAPI
- Uvicorn
- REST API Architecture
- Python-Based Inference Services

The backend coordinates video processing, machine learning inference, and communication between services.

9.3. Computer Vision Technologies

The computer vision layer utilizes:

- MediaPipe Pose
- OpenCV

These technologies enable pose estimation, landmark extraction, frame processing, and visual annotation.

9.4. Machine Learning Technologies

The machine learning pipeline utilizes:

- Scikit-Learn
- NumPy
- Pandas
- Random Forest
- Support Vector Classifier
- Logistic Regression
- K-Nearest Neighbors
- Stochastic Gradient Descent Classifier

10. SYSTEM ARCHITECTURE

PostureIQ follows a layered architecture separating user interaction, business logic, machine learning inference, and analytics generation.

The architecture was designed to ensure modularity, maintainability, and future scalability.

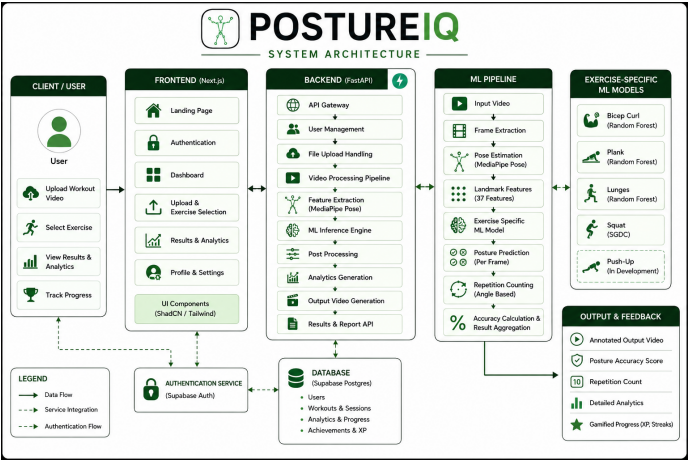


Figure 4. High-Level System Architecture

10.1. System Workflow

The complete workflow follows the sequence:

1. User uploads workout video
2. Frontend sends request to backend
3. Backend processes video frames
4. MediaPipe extracts landmarks
5. Features are generated
6. Exercise-specific model performs inference
7. Posture predictions are generated
8. Repetition counting is executed
9. Accuracy score is calculated
10. Output video is generated
11. Results are displayed on the dashboard

11. FRONTEND ARCHITECTURE

The frontend was designed to provide a modern and intuitive user experience.

Major frontend modules include:

- Landing Page
- Authentication Interface
- Dashboard
- Exercise Selection
- Upload Interface
- Workout History
- Profile Management
- Analytics Visualization

The dashboard acts as the central hub where users can review previous sessions, monitor progress, and analyze workout performance.

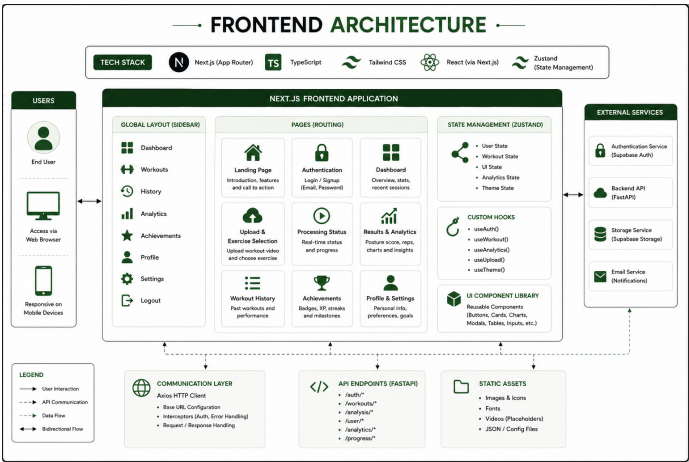


Figure 5. Frontend Architecture Overview

12. BACKEND ARCHITECTURE

The backend was implemented as a modular FastAPI service responsible for exercise analysis and machine learning inference.

Key responsibilities include:

- API Management
- Video Processing
- Landmark Extraction
- Model Inference
- Session Processing
- Analytics Generation
- Output Video Rendering

The backend acts as the bridge between the user interface and the machine learning pipeline.

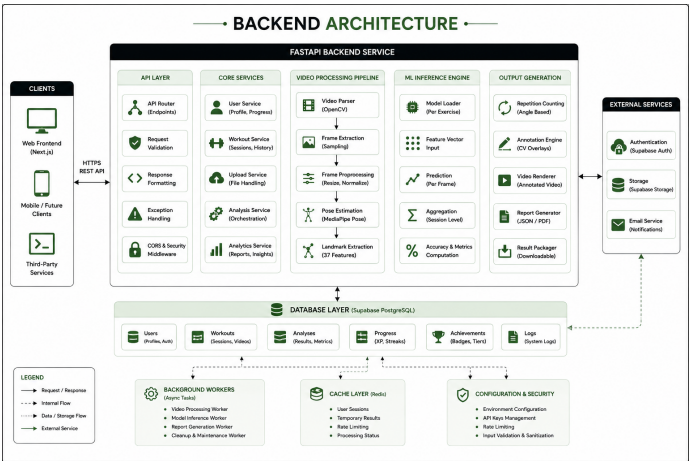


Figure 6. Backend Service Architecture

13. AUTHENTICATION AND DATA PERSISTENCE

Authentication was initially implemented using Supabase services. Supported capabilities included:

- User Registration
- User Login
- Session Management
- Identity Tracking

As the project evolved and features such as XP progression, achievement tiers, workout streaks, and long-term analytics were introduced, the persistence layer required redesign for future scalability.

The current MVP focuses primarily on validating the machine learning workflow while preparing for a more scalable persistence architecture in future releases.

14. MACHINE LEARNING AND COMPUTER VISION PIPELINE

The intelligence layer of PostureIQ is built using a combination of computer vision and machine learning techniques.

The system analyzes exercise videos by extracting pose landmarks and evaluating posture quality through exercise-specific machine learning models.

Unlike generalized exercise classifiers, PostureIQ utilizes dedicated models for individual exercise categories.

15. SUPPORTED EXERCISES

The current version supports:

- Bicep Curl
- Squat
- Lunges
- Plank
- Push-Up (Under Development)

Each exercise uses its own dataset, labels, and machine learning model.

16. COMPUTER VISION PIPELINE

MediaPipe Pose is used to perform pose estimation and landmark extraction.

For every frame:

1. Human pose is detected.
2. Landmarks are extracted.
3. Numerical features are generated.
4. Features are passed to the ML model.
5. Predictions are generated.
6. Session-level analysis is produced.

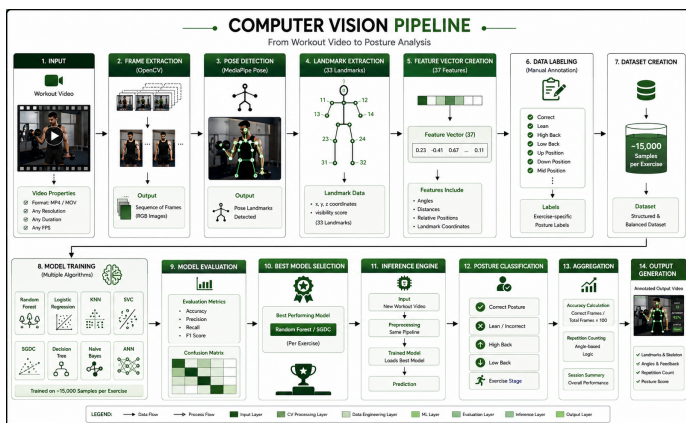


Figure 7. Computer Vision Pipeline

17. DATASET CREATION AND ANNOTATION

One of the most challenging stages of the project was dataset creation. Training data was collected using:

- Self-recorded workout videos
- Public exercise videos
- Open-source exercise analysis references

Videos were converted into frame-level samples and manually annotated.

The final dataset consisted of approximately:

15,000 manually labeled samples per exercise

This process represented the most time-intensive component of the project.

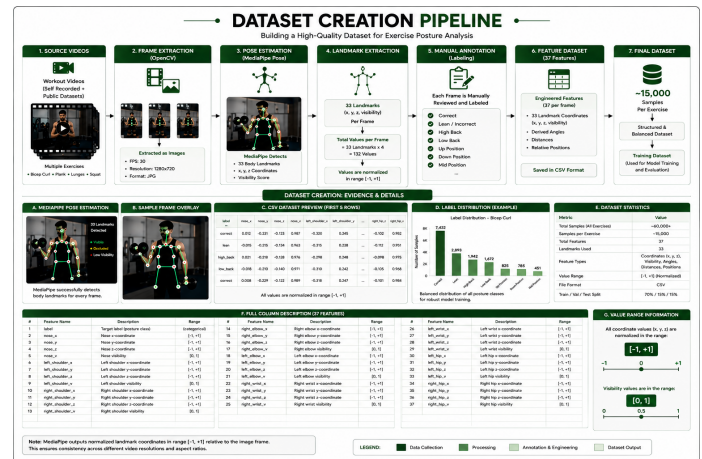


Figure 8. Dataset Collection and Annotation Workflow

18. FEATURE ENGINEERING

Each frame is transformed into a structured feature vector using landmark coordinates extracted from MediaPipe Pose.

The generated dataset contains:

- Approximately 15,000 samples per exercise
- 37 numerical features
- Landmark coordinates
- Visibility measurements
- Exercise-specific labels

These features provide a compact numerical representation of body posture and movement.

19. MACHINE LEARNING MODELS EVALUATED

Multiple machine learning algorithms were trained and compared:

- Random Forest
- Logistic Regression
- K-Nearest Neighbors
- Support Vector Classifier
- Decision Tree
- Naive Bayes
- SGDC
- Deep Neural Networks

Evaluation metrics included:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix Analysis

The best-performing model was selected independently for each exercise category.

20. EXERCISE-SPECIFIC MODELS

PostureIQ utilizes dedicated models for individual exercises.

20.1. Bicep Curl

Model: Random Forest

Labels:

- Correct
- Lean

20.2. Plank

Model: Random Forest
Labels:

- Correct
- Back Too Low
- Back Too High

20.3. Lunges

Model: Random Forest
Labels:

- Initial Position
- Mid Position
- Down Position

20.4. Squat

Model: SGDC
Labels:

- Up
- Down

21. POSTURE ACCURACY CALCULATION

Each frame of an uploaded workout video is independently evaluated using the corresponding exercise-specific machine learning model.
The final posture accuracy score is calculated as:

$$Accuracy = \frac{Correct\ Frames}{Total\ Frames} \times 100 \tag{1}$$

This frame-level aggregation allows PostureIQ to evaluate posture consistency across an entire workout session rather than relying on isolated predictions.

22. REPETITION COUNTING LOGIC

While posture evaluation is performed using machine learning models, repetition counting is implemented using biomechanical angle analysis.

For example, during bicep curl analysis:

- Elbow Angle > 120° → Open State
- Elbow Angle < 20° → Closed State

A complete transition from Open State to Closed State is counted as one repetition.

This deterministic approach was selected due to its reliability, interpretability, and low computational cost.

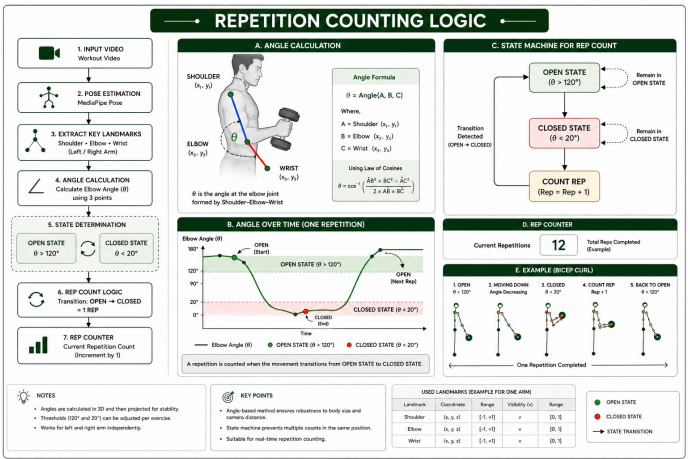


Figure 9. Angle-Based Repetition Counting Logic

23. MODEL EVALUATION AND RESULTS

The performance of PostureIQ was evaluated using exercise-specific datasets and multiple machine learning algorithms.
Models were assessed using:

- Accuracy
- Precision
- Recall
- F1 Score
- Confusion Matrix Analysis

A separate test dataset was used during evaluation to ensure reliable performance measurement.

24. EVALUATION METHODOLOGY

The evaluation workflow consisted of:

1. Dataset Collection
2. Landmark Extraction
3. Manual Annotation
4. Feature Engineering
5. Model Training
6. Validation
7. Testing
8. Performance Comparison

The highest-performing model was selected independently for each exercise category.

25. EXERCISE PERFORMANCE RESULTS

25.1. Bicep Curl

Selected Model: Random Forest

- Accuracy: 93.38
- Precision: 94.72
- Recall: 92.45
- F1 Score: 93.13

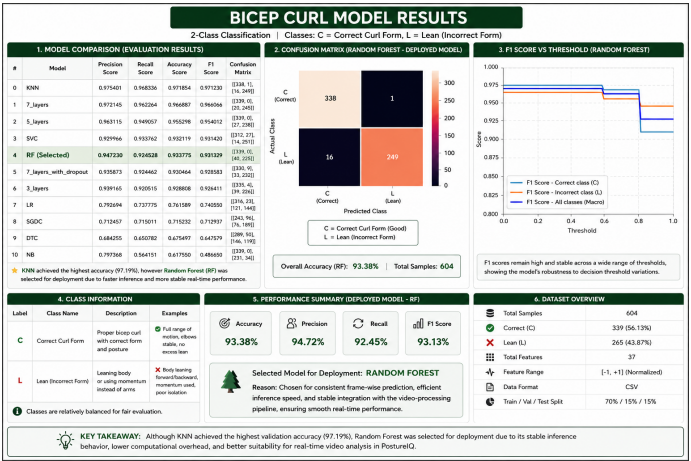


Figure 10. Bicep Curl Evaluation Results

25.2. Plank

Selected Model: Random Forest

- Accuracy: 89.86
- Precision: 92.25
- Recall: 89.86
- F1 Score: 89.62

25.3. Model Selection Decision

Although Logistic Regression achieved the highest validation accuracy (99.58%), Random Forest was selected for deployment.

The deployment decision was based on practical considerations including stable inference behavior, efficient frame-wise processing, and reliable integration with the video analysis pipeline.

This highlights an important engineering principle: the model with the highest validation accuracy is not always the optimal production model.

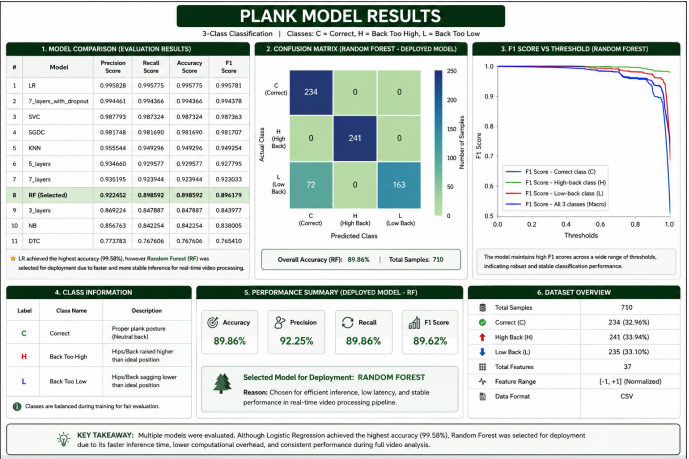


Figure 11. Plank Evaluation Results

25.4. Lunges

Selected Model: Random Forest

- Accuracy: 84.19
- Precision: 85.45
- Recall: 84.19
- F1 Score: 84.07

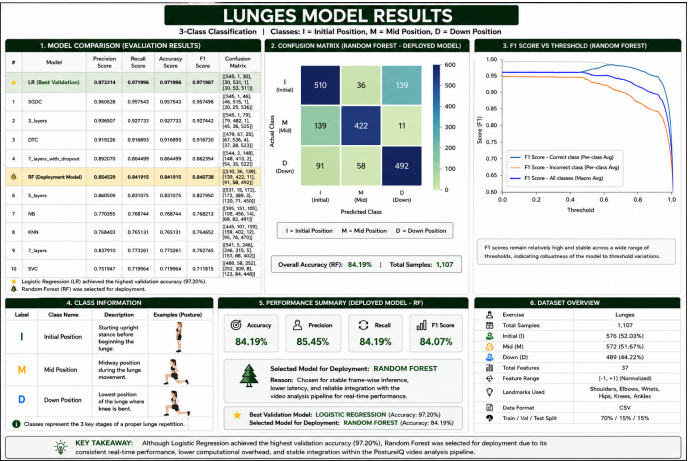


Figure 12. Lunge Evaluation Results

25.5. Squat

Selected Model: SGDC

- Accuracy: 99.30
- Precision: 99.31
- Recall: 99.30
- F1 Score: 99.30

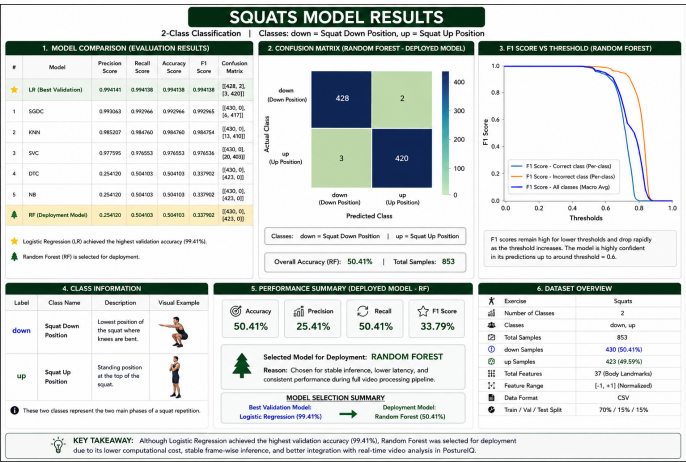


Figure 13. Squat Evaluation Results

26. COMPARATIVE PERFORMANCE SUMMARY

Table 1. Final Model Performance

Exercise	Model	Accuracy
Bicep Curl	Random Forest	93.38%
Plank	Random Forest	89.86%
Lunges	Random Forest	84.19%
Squat	SGDC	99.30%

The results demonstrate that exercise-specific machine learning models provide strong classification performance while maintaining robustness across multiple exercise categories.

27. OUTPUT VISUALIZATION

One of the most valuable features of PostureIQ is its ability to generate annotated output videos.

The generated videos contain:

- Pose Landmarks
- Joint Tracking
- Angle Visualization
- Repetition Count
- Posture Accuracy
- Exercise Analysis Overlays

These visualizations help users understand posture mistakes and movement quality.

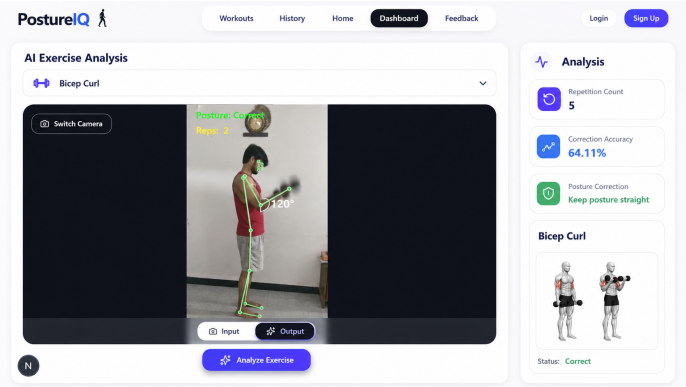


Figure 14. Annotated Output Video Generated by PostureIQ

28. CHALLENGES AND ENGINEERING DECISIONS

28.1. Dataset Collection and Annotation

The most significant challenge encountered during development was creating a high-quality labeled dataset.

Unlike traditional machine learning domains where large public datasets exist, exercise-specific posture datasets are limited.

To overcome this limitation:

- Workout videos were collected from multiple sources.
- Frames were extracted using computer vision pipelines.
- MediaPipe landmarks were generated.
- Samples were manually reviewed and labeled.

Approximately 15,000 samples were labeled for each exercise category.

Key Learning:

Data quality often contributes more to performance than model complexity.

28.2. Real-Time Exercise Analysis

The initial vision of PostureIQ included real-time posture correction using live webcam streams.

However, continuous frame processing introduced significant latency and increased computational requirements.

Engineering Decision:

The team redesigned the MVP around uploaded-video analysis to prioritize stability and user experience.

28.3. Scalability of User Progress Tracking

As the project evolved, additional features such as:

- XP Progression
- Achievement Tiers
- Workout Streaks
- Workout History
- Performance Analytics

increased the complexity of the persistence layer.

The current MVP focuses on validating the machine learning workflow while preparing for a scalable data architecture in future versions.

28.4. Exercise-Specific Movement Patterns

Different exercises exhibit unique posture and movement characteristics.

A single generalized model was insufficient for accurately evaluating all supported exercises.

The solution was to develop independent models for each exercise category.

29. MAJOR ENGINEERING ACHIEVEMENTS

Key project achievements include:

- Full-Stack Fitness Analysis Platform
- Exercise-Specific Machine Learning Models
- Automated Posture Evaluation
- Pose Estimation Pipeline
- Repetition Counting System
- Video Annotation Pipeline
- Dashboard Analytics System
- Gamified User Engagement Features

30. FUTURE SCOPE

Future versions of PostureIQ may include:

- Real-Time Webcam Analysis
- Live Posture Correction
- Push-Up Module Completion
- Automatic Exercise Detection
- Mobile Application Support
- Personalized Workout Recommendations
- Cloud-Based Analytics Infrastructure
- AI-Powered Fitness Coaching
- Injury Risk Prediction

31. CONCLUSION

PostureIQ demonstrates the successful integration of computer vision, machine learning, and full-stack web technologies into a unified fitness analysis platform.

The system enables users to evaluate exercise posture, track repetitions, visualize movement patterns, and receive quantitative performance insights through automated video analysis.

The project involved large-scale dataset creation, manual annotation, model evaluation, backend integration, and user-centric product design. Through exercise-specific machine learning models and computer vision pipelines, PostureIQ achieved strong classification performance while maintaining an intuitive user experience.

Future development will focus on real-time posture correction, cloud scalability, advanced analytics, and intelligent fitness recommendations, transforming PostureIQ from a successful MVP into a comprehensive AI-powered fitness coaching platform.

■ ACKNOWLEDGEMENTS

The authors would like to express their sincere gratitude to the Department of Artificial Intelligence and Data Science for providing the resources, guidance, and academic environment necessary for the successful completion of this project.

We also acknowledge the invaluable contributions of the open-source community, whose tools, frameworks, and research resources played a significant role in the development of PostureIQ.

Special thanks to:

- Google MediaPipe Team
- OpenCV Community
- Scikit-Learn Contributors
- FastAPI Development Team
- Next.js Development Team
- TensorFlow and PyTorch Communities
- Supabase Team
- Open-Source Fitness Analysis Researchers

32. PROJECT REPOSITORY

The complete source code, project documentation, machine learning pipelines, frontend implementation, backend services, and deployment configuration for PostureIQ are available through the project's GitHub repository.

Repository Link:

<https://github.com/MradulBhartiya/PostureIQ>

The repository contains:

- Frontend implementation using Next.js and TypeScript
- Backend APIs developed with FastAPI
- MediaPipe-based pose estimation pipeline
- Machine learning training scripts
- Dataset preprocessing utilities
- Model evaluation and benchmarking code
- Deployment and configuration files

The repository serves as the primary source for project maintenance, future enhancements, and reproducibility of the presented results.

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